



ORBITSENSE

Predictive Space Debris Collision Risk Assessment

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Presentation Agenda

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Problem Statement

Defining the problem & its real-world relevance

02

Literature Review

Key background research & existing solutions

03

Methodology

Technical approach & architecture

04

Implementation

System demo & functional walkthrough

05

Testing & Validation

Results, metrics & QA

06

Innovation & Impact

What makes this unique & future scope

07

Conclusion

Summary, learnings & acknowledgements

Low Earth Orbit Is Running Out of Room

CONTEXT

40,000+ objects are tracked in orbit; only about one third are active satellites. An estimated 1.2 million fragments of 1-10 cm are untracked, yet any could disable a spacecraft.

PROBLEM

Collision-risk intelligence comes from enterprise and government platforms (e.g. LeoLabs). Students, CubeSat teams and academic missions have no open, hands-on tool.

OBJECTIVE

Build an open-data framework: SGP4 physics generates risk features, ML ranks risk, and a live 3D dashboard presents it. Validate against a documented historical collision.

Existing Work and the Gap

Key references

- Hoots & Roehrich (1980), Spacetrack Report No. 3: SGP4 model
- Vallado et al. (2006), Revisiting Spacetrack Report #3: SGP4 implementation
- Clohessy & Wiltshire (1960): relative-motion equations for maneuver modelling
- Liu, Ting & Zhou (2008), Isolation Forest: unsupervised anomaly detection
- NASA ODPO, ESA Space Debris Office, CelesTrak SOCRATES: debris statistics and conjunction data

GAP IDENTIFIED

Enterprise tools are paid and closed; agency work is authoritative but not interactive.

OUR APPROACH

Open data, transparent physics-first method, honest unsupervised ML, built for education.

Physics First, ML Second

1

Data

CelesTrak / Space-Track
TLEs, cached and
validated

2

Propagation

SGP4 for 16,000+
objects

3

Screening

Bucket, coarse, fine
conjunction search

4

Risk

Isolation Forest +
empirical uncertainty +
 P_c

5

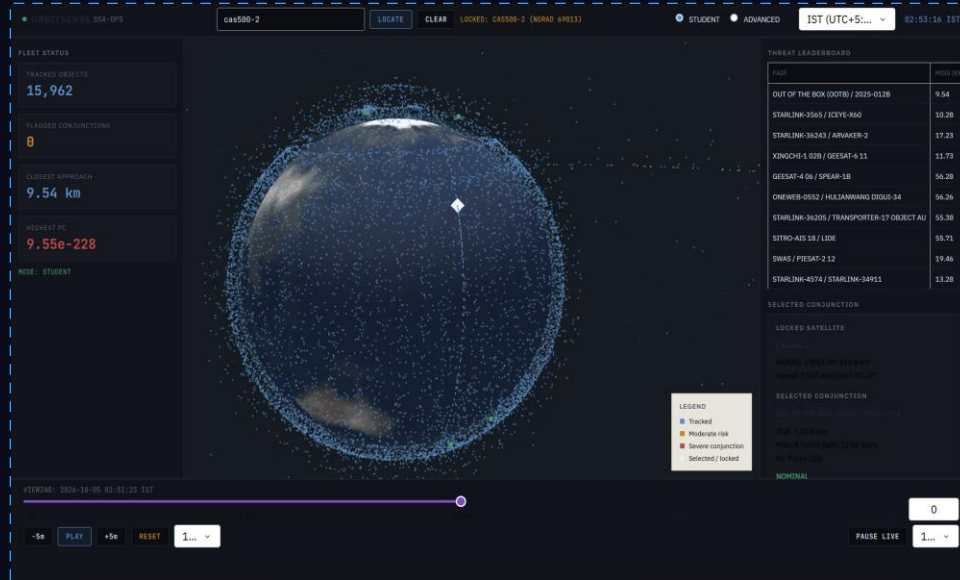
Action

Clohessey-Wiltshire
CAM simulator +
dashboard

TECH STACK

Python, SGP4 / Skyfield, Pandas / NumPy, scikit-learn (Isolation Forest), Plotly / Dash, CelesTrak and Space-Track APIs, GitHub

What Is Built and Running



KEY FEATURES

- Live pipeline: 16,000+ satellites
- Scalable bucket/coarse/fine screening
- Isolation Forest risk ranking
- Empirical uncertainty model + live Pc
- CAM simulator (Clohessy-Wiltshire)
- Physics Sandbox, search, camera lock

GitHub Repository Link: <https://github.com/Shrestha3511/orbitsense>

Reproduced the 2009 Iridium 33 x Cosmos 2251 Collision

699.7 m

OrbitSense miss distance

698.0 m

AGI STK (independent)

1.7 m

Agreement

Validation and CAM results

Relative velocity 11.647 km/s vs documented ~11.7 km/s. A 0.5 m/s along-track burn 2 h before encounter raises miss distance to 8.8 km and drops P_c from $3.9e-05$ to $6.3e-22$.

Bugs found and fixed

Silent 0-satellite fetch (validation + cache); 9.3 km CAM error from dropped fractional second; 50 km Kepler/SGP4 drift (J2) fixed via C-W; dashboard camera reset fixed with Patch().

What Makes This Different

WHAT MAKES THIS UNIQUE

- Independently validated to within 2 m
- Honest ML: anomaly detection, no fabricated labels
- Data-driven uncertainty: $\sigma(t) = 49.3 \sqrt{t}$ m
- Live Pc for every flagged pair
- Interactive Physics Sandbox for learners

FUTURE SCOPE

- Time-scrubber and orbit playback
- 2D ground-track map beside 3D globe
- LEO / MEO / GEO shell filtering
- Collaboration with a CubeSat mission
- Open-source release / publication

Summary

PROJECT SUMMARY

OrbitSense is an open, end-to-end conjunction-risk framework: live SGP4 tracking of 16,000+ objects, ML-assisted ranking, probability of collision and maneuver simulation, validated against a real historical collision.

KEY LEARNINGS

Orbital mechanics, TLE/SGP4, applied ML, teamwork and debugging discipline.

KEY OUTCOMES

Working live dashboard; 1.7 m agreement with AGI STK; CAM simulator; code on GitHub; report and paper.



Team OrbitSense

Awais – Project Manager & Research Lead

Saanvi – Data Acquisition & Orbital Mechanics

Shrestha – Machine Learning & Risk Analysis

Sushennii – Visualization & Dashboard Developer

Bhavya – Testing, Validation & Deployment



Thank You